



PYTHON Exploratory-Data-Analysis:

In []:

IMPORT REQUIRED LIBRARIES:

```
In [14]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud

import warnings
warnings.filterwarnings('ignore')
```

```
In [15]: df = pd.read_csv("netflix_titles.csv")
print("Initial shape:", df.shape)
df.head()
```

Initial shape: (6234, 12)

Out[15]:

	show_id	type	title	director	cast	country	date_added	rele
0	81145628	Movie	Norm of the North: King Sized Adventure	Richard Finn, Tim Maltby	Alan Marriott, Andrew Toth, Brian Dobson, Cole...	United States, India, South Korea, China	September 9, 2019	
1	80117401	Movie	Jandino: Whatever it Takes	NaN	Jandino Asporaat	United Kingdom	September 9, 2016	
2	70234439	TV Show	Transformers Prime	NaN	Peter Cullen, Sumalee Montano, Frank Welker, J...	United States	September 8, 2018	
3	80058654	TV Show	Transformers: Robots in Disguise	NaN	Will Friedle, Darren Criss, Constance Zimmer, ...	United States	September 8, 2018	
4	80125979	Movie	#realityhigh	Fernando Lebrija	Nesta Cooper, Kate Walsh, John Michael Higgins...	United States	September 8, 2017	

DATA CLEANING:

```
In [16]: print("\nMissing values per column:\n")
print(df.isnull().sum())

df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna('Unknown', inplace=True)
df['duration'].fillna('Unknown', inplace=True)
df['date_added'].fillna('Unknown', inplace=True)

df.drop_duplicates(inplace=True)

df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
```

```

# Extract Year and Month from date_added
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month_name()

# Extract numeric duration (for Movies only)
def extract_duration(x):
    try:
        if 'min' in x:
            return int(x.split()[0])
        else:
            return np.nan
    except:
        return np.nan

df['duration_num'] = df['duration'].apply(extract_duration)

df['primary_country'] = df['country'].apply(lambda x: x.split(',')[0] if x !=

```

Missing values per column:

```

show_id      0
type         0
title        0
director    1969
cast        570
country     476
date_added   11
release_year 0
rating       10
duration     0
listed_in    0
description  0
dtype: int64

```

EDA:

```

In [17]: print("\nDataset after cleaning and feature creation:\n")
print(df.info())

```

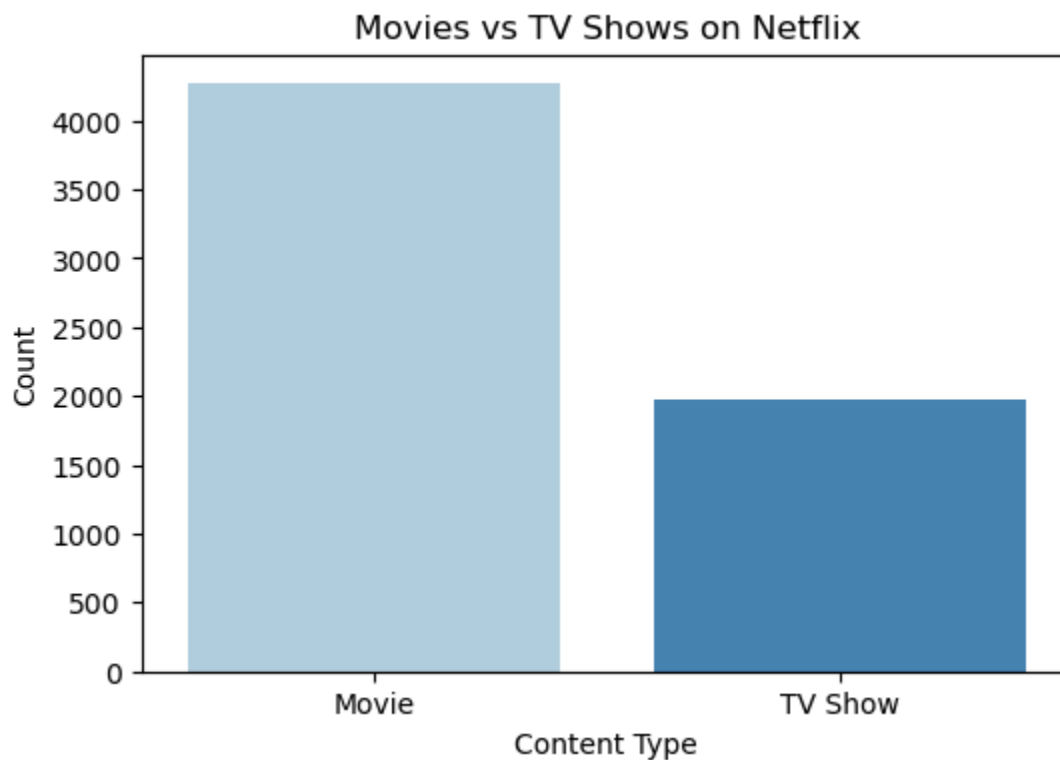
```

# 1. NUMBER OF MOVIES v/s TV Shows:
plt.figure(figsize=(6,4))
sns.countplot(x='type', data=df, palette='Blues')
plt.title('Movies vs TV Shows on Netflix')
plt.xlabel('Content Type')
plt.ylabel('Count')
plt.show()

```

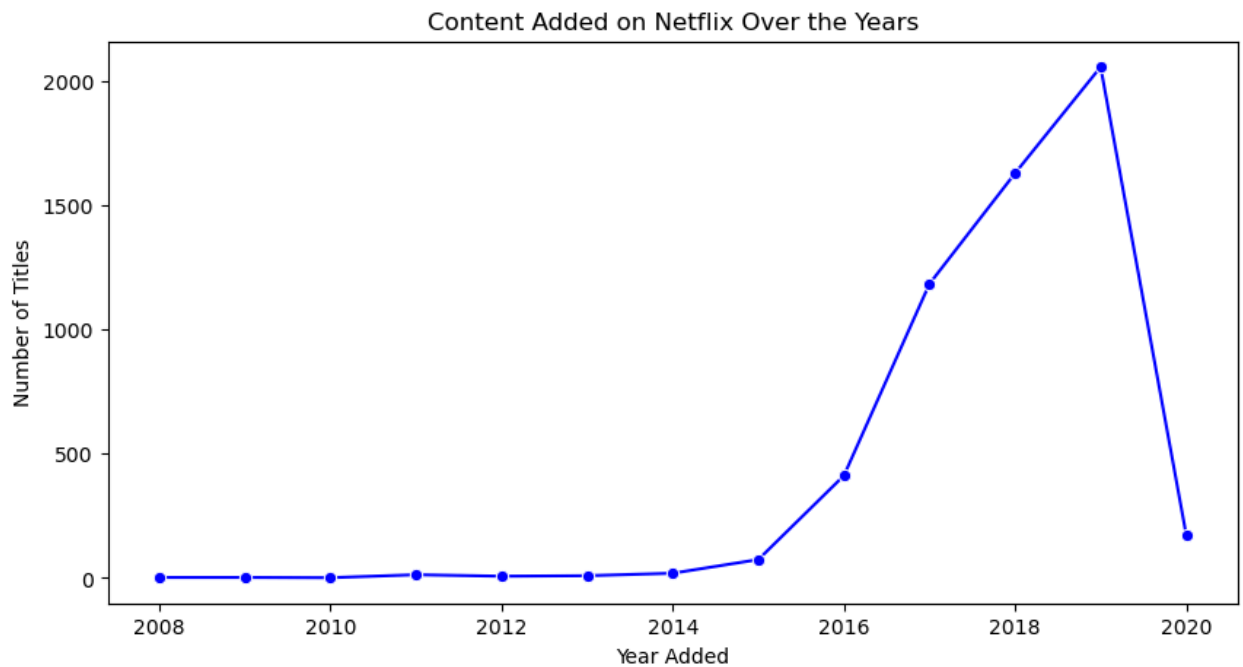
Dataset after cleaning and feature creation:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6234 entries, 0 to 6233
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   show_id               6234 non-null   int64   
1   type                  6234 non-null   object  
2   title                 6234 non-null   object  
3   director              6234 non-null   object  
4   cast                  6234 non-null   object  
5   country               6234 non-null   object  
6   date_added            5583 non-null   datetime64[ns]
7   release_year          6234 non-null   int64   
8   rating                6234 non-null   object  
9   duration              6234 non-null   object  
10  listed_in             6234 non-null   object  
11  description            6234 non-null   object  
12  year_added            5583 non-null   float64  
13  month_added           5583 non-null   object  
14  duration_num          4265 non-null   float64  
15  primary_country       6234 non-null   object  
dtypes: datetime64[ns](1), float64(2), int64(2), object(11)
memory usage: 779.4+ KB
None
```

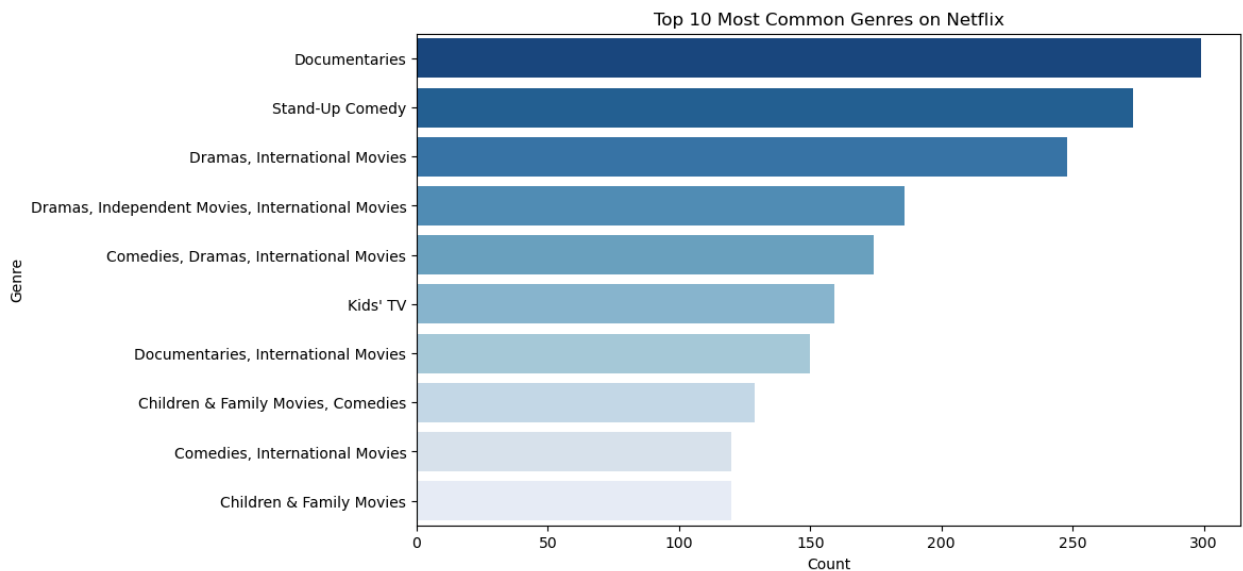


```
In [18]: # 2. YEARLY TRENDS:
yearly_count = df['year_added'].value_counts().sort_index()
plt.figure(figsize=(10,5))
```

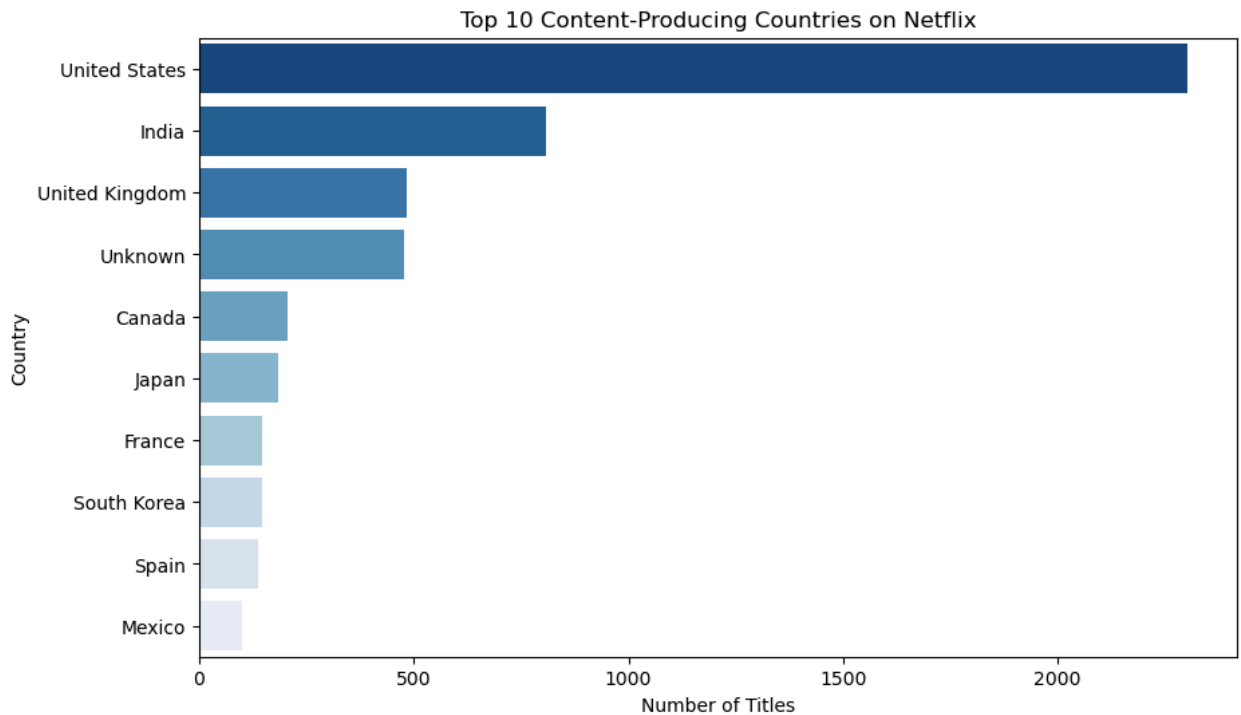
```
sns.lineplot(x=yearly_count.index, y=yearly_count.values, marker='o', color='b')
plt.title('Content Added on Netflix Over the Years')
plt.xlabel('Year Added')
plt.ylabel('Number of Titles')
plt.show()
```



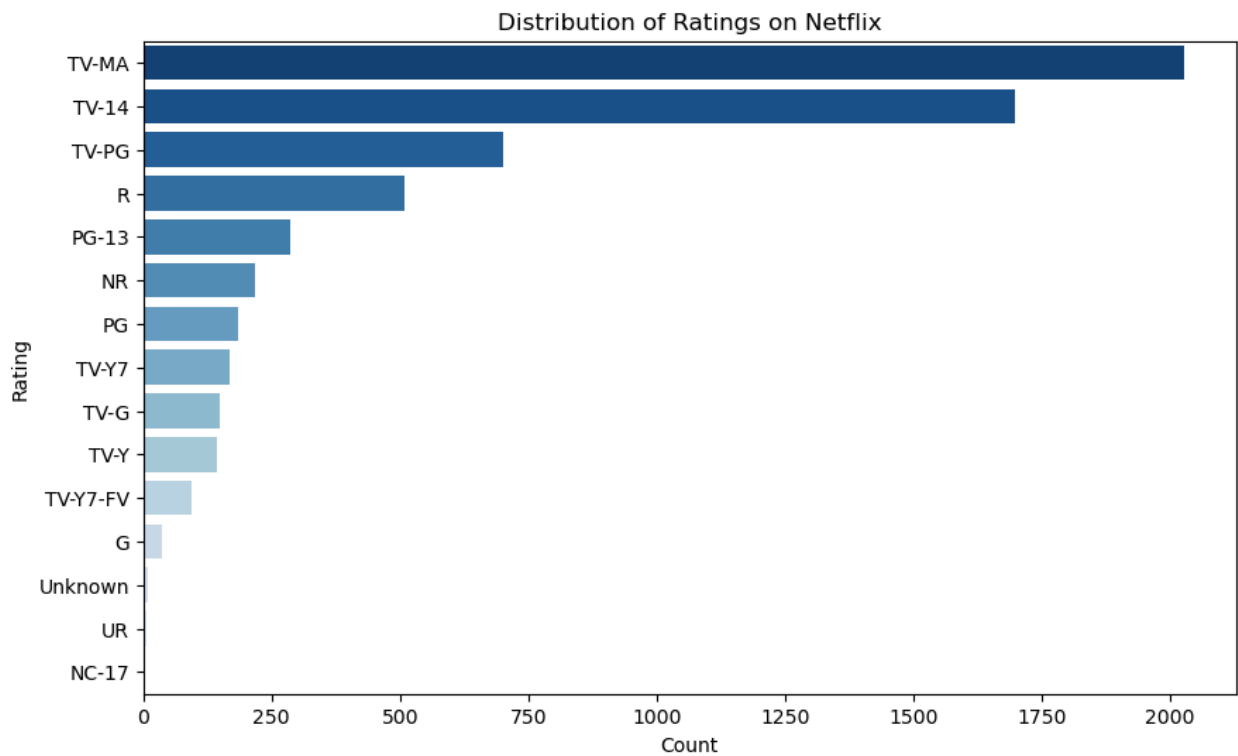
```
In [19]: # 3. HIGH PERFORMANCE GENRES:
plt.figure(figsize=(10,6))
genre_data = df['listed_in'].value_counts().head(10)
sns.barplot(y=genre_data.index, x=genre_data.values, palette='Blues_r')
plt.title('Top 10 Most Common Genres on Netflix')
plt.xlabel('Count')
plt.ylabel('Genre')
plt.show()
```



```
In [20]: # 4. TOP-10 CONTRIBUTING COUNTRIES:
country_data = df['primary_country'].value_counts().head(10)
plt.figure(figsize=(10,6))
sns.barplot(y=country_data.index, x=country_data.values, palette='Blues_r')
plt.title('Top 10 Content-Producing Countries on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Country')
plt.show()
```



```
In [21]: # 5. RATINGS DISTRIBUTION:
plt.figure(figsize=(10,6))
sns.countplot(y='rating', data=df, order=df['rating'].value_counts().index, palette='Blues_r')
plt.title('Distribution of Ratings on Netflix')
plt.xlabel('Count')
plt.ylabel('Rating')
plt.show()
```



```
In [22]: # 6. AVERAGE DURATION OF MOVIES:
avg_duration = df[df['type']=='Movie']['duration_num'].mean()
print(f"\nAverage Movie Duration: {avg_duration:.2f} minutes")
```

Average Movie Duration: 99.10 minutes

```
In [23]: # 7. GENRES WORDCLOUD:
plt.figure(figsize=(10,7))
text = ' '.join(df['listed_in'].dropna().astype(str))
wordcloud = WordCloud(width=1000, height=600, background_color='black', colormap=cm.viridis)
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Most Common Genres on Netflix', fontsize=15)
plt.show()
```

Most Common Genres on Netflix



```
In [24]: df_directors = df.copy()
df_directors["director"] = df_directors["director"].str.split(',')
df_directors=df_directors.explode("director").reset_index(drop=True)
df_directors.head()
```


Out[24]:

	show_id	type	title	director	cast	country	date_added	rele
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0	81145628	Movie	Norm of the North: King Sized Adventure	Richard Finn	Alan Marriott, Andrew Toth, Brian Dobson, Cole...	United States, India, South Korea, China	2019-09-09	
1	81145628	Movie	Norm of the North: King Sized Adventure	Tim Maltby	Alan Marriott, Andrew Toth, Brian Dobson, Cole...	United States, India, South Korea, China	2019-09-09	
2	80117401	Movie	Jandino: Whatever it Takes	Unknown	Jandino Asporaat	United Kingdom	2016-09-09	
3	70234439	TV Show	Transformers Prime	Unknown	Peter Cullen, Sumalee Montano, Frank Welker, J...	United States	2018-09-08	
4	80058654	TV Show	Transformers: Robots in Disguise	Unknown	Will Friedle, Darren Criss, Constance Zimmer, ...	United States	2018-09-08	

In [25]:

```
# 8. DIRECTORS WORDCLOUD:
plt.figure(figsize=(10,7))
text = ' '.join(df_directors['director'].dropna().astype(str))
wordcloud = WordCloud(width=1000, height=600, background_color='black', colormap=cm.viridis)
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Most Common Directors on Netflix', fontsize=15)
plt.show()
```

A word cloud visualization of names, with 'Unknown' being the most prominent word in the center. Other names include David, Chris, Mark, Adam, Martin, Daniel, Michael, Peter, John, Scott, and many others, all in various sizes and colors (blue, green, yellow, orange, red, purple, pink, brown, grey).

```
In [26]: # 9. COUNTRIES-WISE MAP:
try:
    import plotly.express as px
    country_counts = df['primary_country'].value_counts().reset_index()
    country_counts.columns = ['country', 'count']
    fig = px.choropleth(country_counts, locations='country', locationmode='countrynames',
                        color='count', title='Netflix Content by Country',
                        color_continuous_scale='Blues')

    fig.show()
except:
    print("Plotly not installed. Install via: pip install plotly")

# SAVE CLEANED DATA:
df.to_csv("netflix_cleaned.csv", index=False)
print("\n Data cleaning complete. File saved as netflix_cleaned.csv")
```

Data cleaning complete. File saved as netflix_cleaned.csv

In []:

Netflix Titles Dataset — Storytelling & Insights (EDA Report)

1. Introduction - What This Analysis Is About

Netflix hosts thousands of movies and TV shows across genres and countries. In this analysis, we explore:

- What types of content Netflix hosts
- Which countries contribute the most
- Which genres dominate

- How the content library has grown
- Audience rating patterns
- Duration trends
- Popular directors and themes

This storytelling section explains the insights discovered during the Exploratory Data Analysis (EDA) of the **Netflix Titles Dataset (6,234 records)**.

2. Data Overview

The dataset (`netflix_titles.csv`) includes:

- Title, type (Movie/TV Show)
- Director & cast
- Country of origin
- Date added to Netflix
- Release year
- Rating
- Duration
- Listed genres
- Description

Before analysis, the data required cleaning to ensure accuracy.

3. Data Cleaning Summary

Key cleaning steps performed:

** Handling Missing Values**

- Replaced missing values in `director`, `cast`, `country`, `rating`, `duration`, `date_added` with `"Unknown"`.

** Removing Duplicates**

- Duplicate records were removed.

** Converting Columns**

- Converted `date_added` to a proper datetime format.

- Extracted:
 - year_added
 - month_added
- Extracted numeric duration (for movies only).

** Structuring Country Info**

- Created primary_country containing only the first country listed.

After cleaning, the dataset was consistent and ready for exploration.

4. Insights & Storytelling

4.1 Movies vs TV Shows

Netflix hosts more **Movies** than **TV Shows**.

Insight: Movies form the majority of Netflix's library, but TV shows remain vital for keeping users engaged through long-form content.

4.2 Netflix Library Growth Over the Years

Plotting content by year_added shows a major rise between **2015 to 2020**, after which additions stabilize.

Insight: This period aligns with Netflix's global expansion and shift toward producing original content.

4.3 Top Content Genres

The most common genres include:

- International Movies
- Dramas
- Comedies

- Documentaries
- Action & Adventure

Insight: Drama and International content dominate, indicating Netflix's focus on global audiences and diverse storytelling.

4.4 Top Contributing Countries

The top content-producing countries:

1. United States
2. India
3. United Kingdom
4. Canada
5. Japan

Insight: The US is the largest contributor, but India's rising position reflects Netflix's investment in regional markets.

4.5 Ratings Distribution

Most common ratings:

- TV-MA
- TV-14
- TV-PG

Insight: Netflix leans heavily toward **mature content**, which attracts adult audiences globally.

4.6 Average Movie Duration

Average movie duration on Netflix: **~99 minutes**

Insight: Most Netflix movies follow the standard film length of 90-110 minutes.

4.7 Genre Word Cloud

A word cloud generated from `listed_in` highlights:

- Drama
- Comedy
- International
- Documentary
- Action

Insight: These categories reflect Netflix's core content pillars.

4.8 Director Frequency

After splitting director names, a director word cloud reveals recurring creators.

Insight: Certain directors appear frequently, especially in documentaries, comedy specials, and international titles.

4.9 Country-Wise Global View

A choropleth map shows wide global participation, with strong contributions from North America, Europe, and Asia.

Insight: Netflix's catalog has global diversity, aligning with its international user base.

5. Conclusion - What the Data Tells Us

This EDA reveals that Netflix:

- Primarily hosts **Movies**, with steady growth in TV content.
- Expanded rapidly between **2015-2020**.
- Focuses heavily on **Drama**, **Comedy**, and **International** genres.
- Receives major content contributions from the **USA**, but countries like **India** and **UK** are also significant.
- Favors **mature content (TV-MA)**, targeted at adult audiences.
- Offers movies with a typical duration of around **99 minutes**.

- Showcases a diverse range of directors and countries.

Overall, Netflix's strategy is built around **global reach**, **diverse genres**, and **strong movie dominance**, supported by increasing international content offerings.

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